METRIC

MIL-C-51130B
21 June 1983
SUPERSEDING
MIL-C-51130A
17 May 1971

MILITARY SPECIFICATION

CHEMICALS, REAGENT GRADE; GENERAL SPECIFICATION FOR (METRIC)

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers chemicals used as reagents. The purity of these chemicals is determined by melting point, boiling point, specific gravity, or refractive index.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-2020 - Chemicals, Liquid, Dry, and Paste: Packaging Of

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-650 - Explosive; Sampling, Inspection, and Testing

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D86 - Distillation of Petroleum Products
D891 - Specific Gravity, Apparent, of Liquid Industrial Chemicals
D1160 - Distillation of Petroleum Products at Reduced Pressure
D1218 - Refractive Index and Refractive Dispersion of Hydrocarbon Liquids
D3951 - Commercial Packaging

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Physical characteristics. The reagent chemical specified (see 6.1) shall conform to the applicable physical characteristics of table I when tested as specified therein. Melting point shall be determined as specified in 4.2.4.1. Boiling point shall be determined as specified in 4.2.4.2; 95 milliliters (mL) of a 100-mL specimen shall distill between the indicated temperatures. Specific gravity shall be determined as specified in 4.2.4.3 with a tolerance of $\pm$ 0.003. Refractive index shall be determined as specified in 4.2.4.4 with a tolerance of $\pm$ 0.0004.

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TABLE I. Physical characteristics

Reagent chemical	Melting point, °C	Boiling point, °C	Specific gravity at 20°/4°C	Refractive index at 20°C
p-Aminoazobenzene	122 to 123	---	---	---
n-Amyl acetate	---	147 to 149	---	---
tert-Amyl alcohol	---	100 to 103	0.809	1.4052
Anthranilic acid	146 to 147	---	---	---
Benzenesulfonyl chloride	14 to 16	---	---	---
Benzyl acetate	---	92 to 94 (10 mm)	1.054	1.5025
Benzylaniline	34 to 36	---	---	---
Benzylisothiourea hydrochloride	176 to 177	---	---	---
p-Bromoaniline	62 to 64	---	---	---
Bromobenzene	---	154 to 157	1.495	1.5598
p-Bromobenzenesulfonyl chloride	75 to 77	---	---	---
p-Bromobenzoic acid	256 to 258	---	---	---
p-Bromophenacyl bromide	110 to 112	---	---	---
p-Bromophenol	64 to 66	---	---	---
p-Bromophenylhydrazine hydrochloride	199 to 200 (dec)	---	---	---
n-Butyl ether	---	138 to 141	0.768	1.3993
dl-Camphor	173 to 174	---	---	---
a-Chloroacetophenone	54 to 56	---	---	---
Chlorobenzene	---	130 to 132	1.106	1.5246
Chlorosulfonic acid	---	150 to 153	1.760	---

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TABLE I. Physical characterisitcs (continued)

Reagent chemical	Melting point, °C	Boiling point, °C	Specific gravity at 20°/4°C	Refractive index at 20°C
Cinnamic acid	132 to 134	---	---	---
o-Cresolphthalein	223 to 225	---	---	---
Crotonaldehyde	---	100 to 103	0.857	---
Curcumin	179 to 181	---	---	---
Cyclohexanone	---	154 to 156	0.947	1.4500
Diacetyl monoxime	74 to 75	---	---	---
o-Dibutyl phthalate	---	203 to 206 (20 mm)	1.045	1.4924
Dichloramine-T	80 to 82	---	---	---
o-Dichlorobenzene	---	179 to 181	1.305	1.5519
sym-Diethylene glycol monoethyl ether acetate	---	212 to 223	1.0114 (at 20°/20°C)	---
5,5-Dimethyl-1,3-cyclohexanedione	147 to 149	---	---	---
Dimethyldihydroresorcinol	147 to 149	---	---	---
2,2'-Dimethylthiocarbanilide	148 to 152	---	---	---
sym-Di-o-tolylthiourea	75 to 76	---	---	---
3,5-Dinitrobenzoic acid	203 to 205	---	---	---
2,4-Dinitrophenol	112 to 113	---	---	---
Diphenylbenzidine	246 to 248	---	---	---
sym-Diphenylthiourea	153 to 154	---	---	---
Ethylene bromide	---	130.5 to 132.5	2.180	---
Ethylene chlorohydrin	---	127 to 129	1.213	1.4420

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TABLE I. Physical characteristics (continued)

Reagent chemical	Melting point, °C	Boiling point, °C	Specific gravity at 20°/4°C	Refractive index at 20°C
Ethyl phthalate	---	171 to 174 (14 mm)	1.117	1.5018
Hydrazine dihydrochloride	197 to 199	---	---	---
p-Hydroxyazobenzene	155 to 156	---	---	---
Iodine monochloride	26 to 28	---	---	---
Lanthanum nitrate	39.5 to 40.5	---	---	---
Levulose	103 to 105	---	---	---
Mannitol	166 to 168	---	---	---
2-Mercaptobenzothiazole	177 to 180	---	---	---
Monochloroacetic acid	62 to 64	---	---	---
b-Naphthoquinoline	90 to 91	---	---	---
1,4-Naphthoquinone	121 to 124	---	---	---
a-Naphthylamine	49 to 50	---	---	---
b-Naphthylamine	110 to 112	---	---	---
a-Naphthyl isocyanate	---	140 to 142 (12 mm)	1.181	1.6238
5-Nitrobarbituric acid	181 to 183 (dec)	---	---	---
p-Nitrobenzeneazoresorcinol	197 to 198	---	---	---
m-Nitrobenzhydrazide	153 to 155	---	---	---
p-Nitrobenzoic acid	240 to 242	---	---	---
p-Nitrobenzoyl chloride	73 to 75	---	---	---

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TABLE I. Physical characteristics (continued)

Reagent chemical	Melting point, °C	Boiling point, °C	Specific gravity at 20°/4°C	Refractive index at 20°C
p-Nitrobenzyl chloride	71 to 73	---	---	---
Nitromethane	---	100 to 102	1.138	1.3820
p-Nitrophenol	112 to 114	---	---	---
3-Nitrophthalic acid	215(dec)	---	---	---
3-Nitrophthalimide	218 to 220	---	---	---
Phenylarsonic acid	163(dec)	---	---	---
o-Phenylenediamine	101 to 103	---	---	---
p-Phenylenediamine	139 to 141	---	---	---
Phenyl isothiocyanate	---	99 to 100 (15 mm)	1.133	---
1-Phenyl-3-methyl-5-pyrazalone	127 to 128	---	---	---
Phenyl-a-naphthylamine	59.5 to 60.5	---	---	---
p-Phenylphenacyl bromide	124 to 126	---	---	---
1-Phenylsemicarbazide	174 to 176	---	---	---
g-Picoline	---	143 to 145	0.951	1.5050
Picrylsulfonic acid	99 to 101	---	---	---
Pyrocatechol	102 to 104	---	---	---
Salicylaldoxime	57 to 59	---	---	---
p,p'-Tetramethyldiamino-benzophenone	173 to 176	---	---	---
p,p'-Tetramethyldiamino-diphenylmethane	89 to 90	---	---	---

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TABLE I. Physical characteristics (continued)

Reagent chemical	Melting point, °C	Boiling point, °C	Specific gravity at 20°/4°C	Refractive index at 20°C
Thallous acetate	109 to 111	---	---	---
Thiosemicarbazide	180 to 181	---	---	---
p-Toluenesulfonyl chloride	66 to 68	---	---	---
p-Toluidine	43 to 46	---	---	---
2,4,6,-Tribromoaniline	119 to 121	---	---	---
2,4,6-Tribromophenol	93 to 95	---	---	---
Tri-o-cresyl phosphate	---	263 to 266 (20 mm)	1.168	1.5605
Triethylamine	---	88-90	0.728	1.4003
Triiodophenol	156 to 158	---	---	---
1,3,5-Trinitrobenzene	121 to 122	---	---	---
2,4,6-Trinitrobenzoic acid	220(dec)	---	---	---
Triphenyl phosphate	49 to 50	---	---	---
Triphenyltin chloride	105 to 107	---	---	---

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the reagent chemical specified (see 6.1), produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.2).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. See 6.4 for sampling and testing precautions. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 150 grams (g) for solid chemicals and no less than 250 mL for liquid chemicals shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping containers shall be examined for the following defects using an AQL of 1.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container damaged or leaking
(d) Unitization not as specified
(g) Marking incorrect, missing, or illegible

4.2.3.2 For test. See 6.4 for sampling and testing precautions. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be

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thoroughly mixed to form a composited specimen of no less than 150 g for solid chemicals and no less than 250 mL for liquid chemicals. The composited specimen shall be tested as speicifed in the applicable portions of 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. See 6.4 for sampling and testing precautions. Use a well-ventilated fume hood where applicable. Tests shall be conducted as follows:

4.2.4.1 Melting point. Determine melting point in accordance with MIL-STD-650, method 209.1, except that the ground specimen shall be dried for 24 hours over sulfuric acid rather than in an oven.

4.2.4.2 Boiling point.

(a) Atmospheric pressure. Determine the temperatures between which 95 mL of a 100-mL specimen distills in accordance with ASTM D86 and correct to a pressure of 760 millimeters of mercury.

(b) Reduced pressure. Determine the temperature between which 95 mL of a 100-mL specimen distills at the reduced pressure indicated in table I in accordance with ASTM D1160.

4.2.4.3 Specific gravity. Determine specific gravity in accordance with ASTM D891.

4.2.4.4 Refractive index. Determine refractive index in accordance with ASTM D1218.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, C, or industrial as specified (see 6.1).

5.1.1 Levels A, B and C. The specified reagent chemical (see 6.1) shall be preserved in the type and class of unit pack, as specified in PPP-C-2020, and in the quantity as specified (see 6.1). Intermediate packing shall also be in accordance with PPP-C-2020.

5.1.2 Industrial. The specified non-hazardous reagent chemical in the quantity as specified (see 6.1) shall be preserved in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or industrial as specified (see 6.1).

5.2.1 Levels A, B and C. Level A, B and C packing shall be in accordance with PPP-C-2020.

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5.2.2 Industrial packing. Industrial packing shall be in accordance with ASTM D3951.

5.3 Unitization. Level A, B and C packs shall be palletized in accordance with the applicable requirements of PPP-C-2020. Industrial packs shall be unitized in a manner to assure protection of the reagent chemicals during transit and storage and to assure carrier acceptance and stability in stacking.

5.4 Marking. Each container shall be labeled with appropriate statements of hazard and first aid requirements. Level A, B and C containers shall be marked in accordance with PPP-C-2020. Industrial containers shall be marked in accordance with ASTM D3951 and shall also be marked to show the date of manufacture and lot number of the reagent chemical.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) Reagent chemical required (see 3.1),
(c) Level of preservation and packing required (see 5.1 and 5.2), and
(d) Quantity of reagent chemical required (see 5.1.1 and 5.1.2).

6.2 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.3 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.4 Sampling and testing precautions. Some of the chemicals covered by this specification are potentially hazardous to personnel. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of these materials.

Custodians:

Army - EA
Air Force - 68

Review activities:

Army - AR, MD
DLA - GS

User activities:

Army - ME

Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-51130B	CHEMICALS, REAGENT GRADE; GENERAL SPECIFICATION FOR (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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